

The complete Boolean algebra of the Baire space of weight $\aleph_1$:
no nonzero element is $\aleph_1$ -compact

1 The space and its bases

Let

$$X = \omega_1^\omega = \{x \mid x : \omega \rightarrow \omega_1\},$$

carrying the product topology, where ω_1 is given the discrete topology.

Notation for finite sequences

A finite sequence $s \in \omega_1^{<\omega}$ is a function $s : n \rightarrow \omega_1$ for some $n < \omega$; here $n = |s|$ is the *length* of s , i.e. the number of its entries, and (as always in set theory) the ordinal n is the set $\{0, 1, \dots, n-1\}$, so $\text{dom } s = \{0, \dots, |s|-1\}$. We write $s = \langle s(0), \dots, s(|s|-1) \rangle$; in particular $\langle \xi \rangle$ is the sequence of length 1 whose only entry is ξ . Thus $\omega_1^{<\omega} = \bigcup_{n < \omega} \omega_1^n$, and $|\omega_1^{<\omega}| = \aleph_1$.

For a function x and an ordinal $n \subseteq \text{dom } x$ we write $x \upharpoonright n$ for the restriction of x to $\{0, \dots, n-1\}$, and $s \frown t$ for the *concatenation* of s and t : the sequence of length $|s| + |t|$ that runs through s and then through t . So

$$s \frown \langle \xi \rangle = \langle s(0), \dots, s(|s|-1), \xi \rangle, \quad |s \frown \langle \xi \rangle| = |s| + 1,$$

is s with the single extra entry ξ appended. (Many authors abbreviate this as $s \frown \xi$; we keep the angle brackets, since $\frown$ properly takes two sequences.) Finally, $s \subset x$ means that x *end-extends* s , i.e. $x \upharpoonright |s| = s$.

Two bases

The product topology on X admits two standard descriptions. Let $\text{Fn}(\omega, \omega_1)$ denote the set of finite partial functions $p : \omega \rightarrow \omega_1$.

- **Cylinder basis.** For $p \in \text{Fn}(\omega, \omega_1)$ put $C_p = \{x \in X : p \subseteq x\}$: the coordinates in the finite set $\text{dom } p$ are fixed, at *arbitrary* positions, and all others are free.
- **Tree basis.** For $s \in \omega_1^{<\omega}$ put

$$[s] = \{x \in X : s \subset x\} = \{x \in X : x \upharpoonright |s| = s\},$$

the set of all x end-extending s : here the fixed coordinates form an *initial segment* $\{0, \dots, |s|-1\}$ of ω .

The sets $[s]$ are cylinder sets too, namely $[s] = C_s$, but they are special ones. **We work with the tree basis.** This is a convenience, not a necessity (Remark 4.3): the argument needs a free coordinate at which to split a basic set, and for $[s]$ the first free coordinate is simply $|s|$.

(B1) The tree basis is a base of clopen sets. The sets C_p are by definition a base for the product topology. The tree basis *refines* it: given $x \in C_p$, let $n = \max(\text{dom } p) + 1$ (with $n = 0$ if $p = \emptyset$) and let $s = x \upharpoonright n$. Then $s \subset x$, so $x \in [s]$, and every $y \in [s]$ agrees with x on $\text{dom } p$, so $p \subseteq y$ and hence $[s] \subseteq C_p$. Thus for every open O and every $x \in O$ there is s with $x \in [s] \subseteq O$, which is exactly what it means for $\{[s] : s \in \omega_1^{<\omega}\}$ to be a base.

Each $[s]$ is open, and its complement

$$X \setminus [s] = \bigcup \{[t] : t \in \omega_1^{|s|}, t \neq s\}$$

is open as well, so $[s]$ is clopen. In particular X is zero-dimensional. (The general cylinders are clopen too, since $X \setminus C_p = \bigcup \{C_q : \text{dom } q = \text{dom } p, q \neq p\}$.)

(B1') The metric picture. Define $d(x, x) = 0$ and $d(x, y) = 2^{-\min\{n : x(n) \neq y(n)\}}$ for $x \neq y$. This is a complete ultrametric inducing the product topology, and the *tree-basis* sets are precisely its balls: $[s]$ is the ball of radius $2^{-|s|}$ about any of its points. (A general cylinder, say one fixing only the coordinate 5, is not a ball.)

(B2) $w(X) = \aleph_1$. There are exactly $\aleph_1$ sequences in $\omega_1^{<\omega}$, so by (B1) $w(X) \leq \aleph_1$. Conversely, the $\aleph_1$ pairwise disjoint nonempty open sets $[\langle \xi \rangle]$, $\xi < \omega_1$, force every base to have size at least $\aleph_1$. Thus X is *the* Baire space of weight $\aleph_1$, often denoted $B(\aleph_1)$.

2 Two lattices of open sets

We consider two complete lattices built from X , both ordered by $\subseteq$.

- $\mathcal{O}(X)$, the lattice of *all* open subsets of X . Here $\bigvee_i U_i = \bigcup_i U_i$.
- $\text{RO}(X) = \{U \subseteq X \text{ open} : U = \text{int cl } U\}$, the regular open sets. This is a complete Boolean algebra with $U \wedge W = U \cap W$, $\neg U = X \setminus \text{cl } U$, and

$$\bigvee_{i \in I} U_i = \text{int cl} \left(\bigcup_{i \in I} U_i \right).$$

Although $\text{RO}(X) \subseteq \mathcal{O}(X)$ with the same order, joins in the two lattices differ, so a property of an element defined via joins (such as compactness, Section 3) may depend on which lattice it is taken in. Three facts about $\text{RO}(X)$ are used below.

Lemma 2.1. *If $U \in \text{RO}(X)$ and $C \subseteq X$ is clopen, then $U \cap C \in \text{RO}(X)$.*

Proof. $U \cap C$ is open, hence $U \cap C \subseteq \text{int cl}(U \cap C)$. Conversely, C is closed, so $\text{cl}(U \cap C) \subseteq \text{cl}(U) \cap C$, and therefore

$$\text{int cl}(U \cap C) \subseteq \text{int}(\text{cl}(U) \cap C) = \text{int cl}(U) \cap \text{int}(C) = U \cap C,$$

using that C is open in the last step. □

Lemma 2.2. *If $U_i \in \text{RO}(X)$ for $i \in I$ and $\bigcup_{i \in I} U_i$ is itself regular open, then $\bigvee_{i \in I} U_i = \bigcup_{i \in I} U_i$ in $\text{RO}(X)$.*

Lemma 2.3. *If $U_i \subseteq W$ for all $i \in I$ and $W \in \text{RO}(X)$, then $\bigvee_{i \in I} U_i \subseteq W$ in $\text{RO}(X)$.*

Proof. int cl is monotone and $\text{int cl } W = W$. □

3 $\aleph_1$ -compactness and the regularity of ω_1

Lemma 3.1 (regularity of ω_1). *If $A \subseteq \omega_1$ is countable, then $\sup A < \omega_1$.*

Proof. If $A = \emptyset$ then $\sup A = 0$. Otherwise $\sup A = \bigcup A$ is an ordinal, and each $\alpha \in A$ is countable, since ω_1 is the set of countable ordinals. Enumerate $A = \{\alpha_n : n < \omega\}$ (repetitions allowed) and *choose* for each nonzero α_n a surjection $e_n : \omega \rightarrow \alpha_n$. Then $(n, m) \mapsto e_n(m)$ maps a subset of $\omega \times \omega$ onto $\bigcup A$. So $\bigcup A$ is a countable ordinal, hence $\bigcup A < \omega_1$. $\square$

The proof makes countably many arbitrary choices, so it uses the axiom of countable choice, and this cannot be avoided: ZF alone does not prove Lemma 3.1. In the Feferman–Lévy model, ω_1 is a countable union of countable ordinals, so $\text{cf } \omega_1 = \omega$. The same use of choice gives the more general fact that a countable union of countable sets is countable, which is used in Lemma 3.3. We work in ZFC throughout.

Definition 3.2. Let L be a complete lattice and κ an infinite cardinal.

- (a) $a \in L$ is κ -compact if whenever $a \leq \bigvee S$ for some $S \subseteq L$, there is $S_0 \subseteq S$ with $|S_0| < \kappa$ and $a \leq \bigvee S_0$.
- (b) L is κ -algebraic if every element of L is a join of κ -compact elements.

A set $D \subseteq L$ is $\aleph_1$ -directed if it is nonempty and every countable subset of D has an upper bound in D .

For $\kappa = \aleph_1$ there are two common formulations of compactness:

- (i) whenever $a \leq \bigvee S$, some countable $S_0 \subseteq S$ has $a \leq \bigvee S_0$ (this is Definition 3.2(a));
- (ii) whenever $a \leq \bigvee D$ with D $\aleph_1$ -directed, then $a \leq d$ for some $d \in D$.

Lemma 3.3. *For a in a complete lattice L , (i) and (ii) are equivalent.*

Proof. (i) $\Rightarrow$ (ii): given $a \leq \bigvee D$ with D $\aleph_1$ -directed, (i) yields a countable $D_0 \subseteq D$ with $a \leq \bigvee D_0$, and an upper bound $d \in D$ of D_0 satisfies $a \leq \bigvee D_0 \leq d$.

(ii) $\Rightarrow$ (i): given $a \leq \bigvee S$, let $D = \{\bigvee S_0 : S_0 \subseteq S \text{ countable}\}$. Then $D \subseteq S$ and D consists of elements $\leq \bigvee S$, so $\bigvee D = \bigvee S$. And D is $\aleph_1$ -directed: given countably many $d_n \in D$, choose countable $S_n \subseteq S$ with $d_n = \bigvee S_n$; then $\bigcup_n S_n$ is countable (here we use countable choice, as in Lemma 3.1), and $\bigvee \bigcup_n S_n \in D$ bounds every d_n . By (ii), $a \leq \bigvee S_0$ for some countable $S_0 \subseteq S$. $\square$

4 The chain below a nonzero element

Proposition 4.1. *Let $U \in \text{RO}(X)$ be nonzero. Then there is a chain $\langle U_\alpha : \alpha < \omega_1 \rangle$ in $\text{RO}(X)$ such that:*

- (1) $U_\alpha \subsetneq U_\beta$ whenever $\alpha < \beta < \omega_1$;
- (2) $U_\alpha \subsetneq U$ for every $\alpha < \omega_1$;
- (3) $U_\lambda = \bigvee_{\alpha < \lambda} U_\alpha$ for every limit $\lambda < \omega_1$ (the chain is continuous);

$$(4) \bigvee_{\alpha < \omega_1} U_\alpha = U;$$

$$(5) \bigvee_{\alpha \in A} U_\alpha \neq U \text{ for every countable } A \subseteq \omega_1.$$

Proof. Step 1: a basic open set inside U .

U is nonempty and open, so by (B1) there is $s \in \omega_1^{<\omega}$ with

$$V := [s] \subseteq U.$$

The inclusion $V \subseteq U$ is what makes the rest work; it is used in Steps 3, 4 and 6.

Step 2: splitting V at the next coordinate.

Since $V = [s]$ fixes exactly the coordinates $0, \dots, |s| - 1$, the first free coordinate of a point of V is the one with index $|s|$. For $\xi < \omega_1$ put

$$V_\xi = [s \frown \langle \xi \rangle] = \{x \in X : x \restriction |s| = s \text{ and } x(|s|) = \xi\},$$

taking *all* $\xi < \omega_1$; in tree language, the V_ξ are the immediate successors of the node s . Then

$$V = \prod_{\xi < \omega_1} V_\xi,$$

a partition of V into $\aleph_1$ nonempty clopen pieces. Indeed:

- each $V_\xi \subseteq V$, since $s \subset s \frown \langle \xi \rangle \subset x$ for $x \in V_\xi$;
- they cover V : if $x \in V$ then x is a function, so $\xi := x(|s|)$ is a single well-defined ordinal $< \omega_1$, and $x \in V_\xi$;
- they are pairwise disjoint: an $x \in V_\xi \cap V_\eta$ would have $\xi = x(|s|) = \eta$;
- each is nonempty: V_ξ contains the point $s \frown \langle \xi \rangle \frown \langle 0, 0, 0, \dots \rangle$.

This splitting is done *once*. The chain below has length ω_1 and runs over the pieces V_ξ , not over coordinates.

Step 3: the chain.

For $\alpha < \omega_1$ set

$$U_\alpha = (U \setminus V) \cup \bigcup_{\xi < \alpha} V_\xi.$$

Let $W_\alpha = (X \setminus V) \cup \bigcup_{\xi < \alpha} V_\xi$. Then W_α is open, and its complement $\bigcup_{\alpha \leq \xi < \omega_1} V_\xi$ is open too, so W_α is clopen. Since $V_\xi \subseteq V \subseteq U$ for all ξ ,

$$U \cap W_\alpha = (U \setminus V) \cup \bigcup_{\xi < \alpha} V_\xi = U_\alpha,$$

so $U_\alpha \in \text{RO}(X)$ by Lemma 2.1. (For $\alpha = 0$ this reads $U_0 = U \setminus V$, which is 0 in case $U = V$; this is harmless.)

Step 4: strictly increasing, and strictly below U .

Monotonicity in α is immediate from the definition. For strictness, fix $\alpha < \omega_1$ and note that V_α is nonempty, $V_\alpha \subseteq V \subseteq U$, and

$$V_\alpha \cap U_\alpha = \emptyset,$$

because $V_\alpha \cap (U \setminus V) = \emptyset$ and $V_\alpha \cap V_\xi = \emptyset$ for $\xi < \alpha$. Hence $U_\alpha \subsetneq U$, which is (2); and for $\alpha < \beta$ we have $V_\alpha \subseteq U_\beta \setminus U_\alpha$, so $U_\alpha \subsetneq U_\beta$, which is (1). As these are distinct sets lying in $\text{RO}(X)$, they are distinct elements of the algebra.

Step 5: continuity.

Let $\lambda < \omega_1$ be a limit ordinal. Then $\bigcup_{\alpha < \lambda} \bigcup_{\xi < \alpha} V_\xi = \bigcup_{\xi < \lambda} V_\xi$, so

$$\bigcup_{\alpha < \lambda} U_\alpha = (U \setminus V) \cup \bigcup_{\xi < \lambda} V_\xi = U_\lambda,$$

which is regular open by Step 3. Lemma 2.2 then gives $\bigvee_{\alpha < \lambda} U_\alpha = U_\lambda$, which is (3).

Step 6: the join is U .

Using $V \subseteq U$ and Step 2,

$$\bigcup_{\alpha < \omega_1} U_\alpha = (U \setminus V) \cup \bigcup_{\xi < \omega_1} V_\xi = (U \setminus V) \cup V = U.$$

The union equals U , which is regular open, so by Lemma 2.2 the join equals the union: $\bigvee_{\alpha < \omega_1} U_\alpha = U$. This is (4). Note that Lemma 2.2 is genuinely needed here: in general a join in $\text{RO}(X)$ is int cl of the union and may be strictly larger than the union.

Step 7: countable subfamilies fail.

Let $A \subseteq \omega_1$ be countable. By Lemma 3.1, $\beta := \sup A < \omega_1$. The chain is increasing, so $U_\alpha \subseteq U_\beta$ for every $\alpha \in A$, and $U_\beta \in \text{RO}(X)$; hence by Lemma 2.3

$$\bigvee_{\alpha \in A} U_\alpha \subseteq U_\beta \subsetneq U,$$

the last inclusion by Step 4. This is (5). The same computation shows that $D = \{U_\alpha : \alpha < \omega_1\}$ is $\aleph_1$ -directed: the countable subset $\{U_\alpha : \alpha \in A\}$ is bounded by $U_{\sup A} \in D$. $\square$

Corollary 4.2. *No nonzero element of $\text{RO}(X)$ is $\aleph_1$ -compact. Hence the only $\aleph_1$ -compact element of $\text{RO}(X)$ is 0, and $\text{RO}(X)$ is not $\aleph_1$ -algebraic.*

Proof. Let $U \neq 0$ and let $D = \{U_\alpha : \alpha < \omega_1\}$ be as in Proposition 4.1. Formulation (ii) fails for U : D is $\aleph_1$ -directed (Step 7), $U = \bigvee D$ by (4), and $U \not\subseteq U_\alpha$ for every α by (2). Formulation (i) fails as well, directly: take $S = D$; by (5), every countable $S_0 \subseteq S$ has $\bigvee S_0 \subsetneq U$. Finally, every join of $\aleph_1$ -compact elements is then 0, while the top element X is not 0. $\square$

Remark 4.3 (Why the tree basis). Either basis works. Given a cylinder $C_p \subseteq U$, choose any $n \notin \text{dom } p$ (the least one, say) and put $V_\xi = C_{p \cup \{(n, \xi)\}}$; these partition C_p into $\aleph_1$ nonempty clopen pieces, and Steps 3–7 go through word for word. The tree basis merely makes the choice of n automatic: for $[s]$ it is $n = |s|$. The set W_α of Step 3 is not basic in either sense, only clopen, so no other feature of the basis is used.

5 A shorter route: antichains

The chain is not actually needed for Corollary 4.2; the partition of Step 2 suffices. This is an instance of a general fact about complete Boolean algebras. Recall that an *antichain* in a Boolean algebra B is a set of nonzero, pairwise disjoint elements, and that for $a \in B$, $B \restriction a$ is *ccc* if every antichain below a is countable. Recall also that every complete Boolean algebra satisfies the infinite distributive law $b \wedge \bigvee S = \bigvee_{s \in S} (b \wedge s)$.

Proposition 5.1. *Let B be a complete Boolean algebra and $a \in B$. Then a is $\aleph_1$ -compact if and only if $B \restriction a$ is ccc.*

Proof. ($\Leftarrow$) Suppose $a \leq \bigvee S$. Let E be the set of nonzero $e \leq a$ such that $e \leq s$ for some $s \in S$, and let $\mathcal{A} \subseteq E$ be a maximal antichain (Zorn's lemma). By hypothesis $\mathcal{A}$ is countable. For each $e \in \mathcal{A}$ choose $s_e \in S$ with $e \leq s_e$, and let $S_0 = \{s_e : e \in \mathcal{A}\}$, a countable set. Put $c = a \wedge \neg \bigvee S_0$ and suppose $c \neq 0$. Since $c \leq a \leq \bigvee S$, distributivity gives $c = \bigvee_{s \in S} (c \wedge s)$, so $c' := c \wedge s \neq 0$ for some $s \in S$. Then $c' \in E$, and c' is disjoint from every $e \in \mathcal{A}$, because $e \leq \bigvee S_0$ while $c' \leq \neg \bigvee S_0$; so $\mathcal{A} \cup \{c'\}$ is a strictly larger antichain in E , contradicting maximality. Hence $c = 0$, i.e. $a \leq \bigvee S_0$.

($\Rightarrow$) Suppose $\mathcal{A}$ is an uncountable antichain below a , and put $c = a \wedge \neg \bigvee \mathcal{A}$ and $S = \mathcal{A} \cup \{c\}$. Since $\bigvee \mathcal{A} \leq a$, we get $\bigvee S = c \vee \bigvee \mathcal{A} = a$. Let $S_0 \subseteq S$ be countable and pick $b \in \mathcal{A} \setminus S_0$. Every member of S_0 is disjoint from b (for c , because $b \leq \bigvee \mathcal{A}$), so by distributivity $b \wedge \bigvee S_0 = 0$. As $0 \neq b \leq a$, this gives $a \not\leq \bigvee S_0$. $\square$

Corollary 5.2. *Every nonzero $U \in \text{RO}(X)$ has an antichain $\{V_\xi : \xi < \omega_1\}$ below it (Step 2), so $\text{RO}(X)$ is nowhere ccc, and Corollary 4.2 follows from Proposition 5.1 without Steps 3–7. In fact, by Proposition 5.1, for any complete Boolean algebra “nowhere ccc” is equivalent to “no nonzero element is $\aleph_1$ -compact.”*

Remark 5.3 (The chain as a translation). Steps 3–7 are an instance of a general construction that turns an uncountable antichain into a witness for formulation (ii). If $a = c \vee \bigvee_{\xi < \omega_1} b_\xi$ where $\{c\} \cup \{b_\xi : \xi < \omega_1\}$ are pairwise disjoint and every $b_\xi \neq 0$, then $a_\alpha = c \vee \bigvee_{\xi < \alpha} b_\xi$ satisfies (1)–(5) with a in place of U ; the verification is Steps 4–7 with Lemmas 2.1–2.3 replaced by the distributive law. In $\text{RO}(X)$ this is the case $c = U \setminus V$, $b_\xi = V_\xi$.

Remark 5.4 (Choice). The ($\Rightarrow$) direction of Proposition 5.1 uses neither choice nor Lemma 3.1: it only needs uncountably many pieces. Applied to the partition of Step 2 it shows, in ZF alone, that no nonzero element of $\text{RO}(X)$ is $\aleph_1$ -compact in sense (i). By contrast, property (5) of the chain depends on Lemma 3.1 and can fail without choice: if $A \subseteq \omega_1$ is countable and cofinal, as in the Feferman–Lévy model, then $\bigcup_{\alpha \in A} U_\alpha = U$.

6 The lattice of all open sets

In $\mathcal{O}(X)$ joins are unions, so $U \in \mathcal{O}(X)$ is $\aleph_1$ -compact if and only if every cover of U by open sets has a countable subcover, i.e. if and only if U is Lindelöf.

Proposition 6.1. *No nonempty open subset of X is Lindelöf. Equivalently, no nonempty $U \in \mathcal{O}(X)$ is $\aleph_1$ -compact in $\mathcal{O}(X)$.*

Proof. Let U be nonempty and open, and let $V = [s] \subseteq U$ and V_ξ be as in Steps 1–2. The family

$$\mathcal{C} = \{U \setminus V\} \cup \{V_\xi : \xi < \omega_1\}$$

consists of open sets (since V is closed), which are pairwise disjoint and whose union is exactly U . A point of V_ξ lies in no other member of $\mathcal{C}$, so every subcover of $\mathcal{C}$ contains all $\aleph_1$ of the sets V_ξ . Hence $\mathcal{C}$ has no countable subcover. $\square$

The chain of Proposition 4.1 also works verbatim in $\mathcal{O}(X)$, for formulation (ii): the U_α are open, $D = \{U_\alpha\}$ is $\aleph_1$ -directed, $\bigcup_\alpha U_\alpha = U$, and $U \not\subseteq U_\alpha$. Since joins in $\mathcal{O}(X)$ are unions, Lemmas 2.1–2.3 are not needed there.

Corollary 6.2. *The only $\aleph_1$ -compact element of $\mathcal{O}(X)$ is $\emptyset$. In particular no nonempty regular open set is $\aleph_1$ -compact in $\mathcal{O}(X)$, and $\mathcal{O}(X)$ is not $\aleph_1$ -algebraic.*

Remark 6.3 (The two notions are independent in general). For a general space Y and $U \in \text{RO}(Y)$, being $\aleph_1$ -compact in $\mathcal{O}(Y)$ means that U is Lindelöf, while by Proposition 5.1 being $\aleph_1$ -compact in $\text{RO}(Y)$ means that $\text{RO}(Y) \upharpoonright U$ is ccc. They come apart because a family can have join U in $\text{RO}(Y)$ without covering U : its union need only be dense in U . Two examples:

- Let $Y = D \cup \{\infty\}$ be the one-point compactification of an uncountable discrete set D . Then Y is compact, so the top element is compact in $\mathcal{O}(Y)$. But $U \mapsto U \cap D$ is an isomorphism $\text{RO}(Y) \cong \mathcal{P}(D)$, in which the top element is not $\aleph_1$ -compact (it is the join of the singletons).
- Let Y be the Sorgenfrey plane. It is separable, hence ccc, so every element of $\text{RO}(Y)$ is $\aleph_1$ -compact. But the antidiagonal is an uncountable closed discrete subset, so Y is not Lindelöf, and the top element is not $\aleph_1$ -compact in $\mathcal{O}(Y)$.

What makes both notions fail together for $X = \omega_1^\omega$ is that every nonempty open set has an uncountable partition into open pieces that cover it *exactly*, with no boundary left over. The clopen tree basis is what provides this.

7 Sharpness: $\aleph_2$

Proposition 7.1. *Every element of $\mathcal{O}(X)$ is $\aleph_2$ -compact in $\mathcal{O}(X)$, and every element of $\text{RO}(X)$ is $\aleph_2$ -compact in $\text{RO}(X)$. Hence both lattices are $\aleph_2$ -algebraic, and neither is $\aleph_1$ -algebraic.*

Proof. Let S be a family of open sets (in either lattice; regular open sets are open). Let T be the set of $t \in \omega_1^{<\omega}$ such that $[t] \subseteq s$ for some $s \in S$, and for each $t \in T$ choose such an $s_t \in S$. Then $S_0 = \{s_t : t \in T\}$ has size at most $|\omega_1^{<\omega}| = \aleph_1$, and $\bigcup S_0 = \bigcup S$: if $x \in s \in S$, then by (B1) there is t with $x \in [t] \subseteq s$, so $t \in T$ and $x \in [t] \subseteq s_t$. In both lattices the join of a family depends only on its union (it is the union in $\mathcal{O}(X)$, and int cl of the union in $\text{RO}(X)$), so $\bigvee S_0 = \bigvee S$. Hence $a \leq \bigvee S$ implies $a \leq \bigvee S_0$. The last sentence follows from Corollaries 4.2 and 6.2. $\square$

So $\aleph_1$ is exactly the threshold at which compactness fails, in both lattices.

8 Further remarks

Remark 8.1 (Identification of the algebra; the two bases as two posets). By (B1) and Step 1, the map $s \mapsto [s]$ is an order isomorphism from $(\omega_1^{<\omega}, \supseteq)$ onto a dense subset of $\text{RO}(X) \setminus \{0\}$, and likewise $p \mapsto C_p$ for $(\text{Fn}(\omega, \omega_1), \supseteq)$. So the two bases of Section 1 are the topological shadow of two forcing posets, and the refinement argument in (B1) is precisely the statement that $\omega_1^{<\omega}$ is dense in $\text{Fn}(\omega, \omega_1)$: every finite partial function extends to one whose domain is an initial segment. Dense subposets have isomorphic completions, whence

$$\text{RO}(X) \cong \text{RO}(\omega_1^{<\omega}) \cong \text{RO}(\text{Coll}(\omega, \omega_1)),$$

the Boolean algebra of the Lévy collapse of ω_1 to ω .

Remark 8.2 ($\text{RO}(X)$ as the Booleanization of $\mathcal{O}(X)$). $\mathcal{O}(X)$ is a frame (a complete Heyting algebra) whose pseudocomplement is $\neg U = \text{int}(X \setminus U) = X \setminus \text{cl} U$, so $\neg \neg U = \text{int cl} U$, and $\text{RO}(X)$ is exactly the set of fixed points of $\neg \neg$: the Booleanization of $\mathcal{O}(X)$. Passing from $\text{RO}(X)$ to $\mathcal{O}(X)$ loses complementation, which is why forcing works with $\text{RO}(X)$; for the compactness questions of this note it loses nothing, and the proofs in $\mathcal{O}(X)$ are slightly simpler.